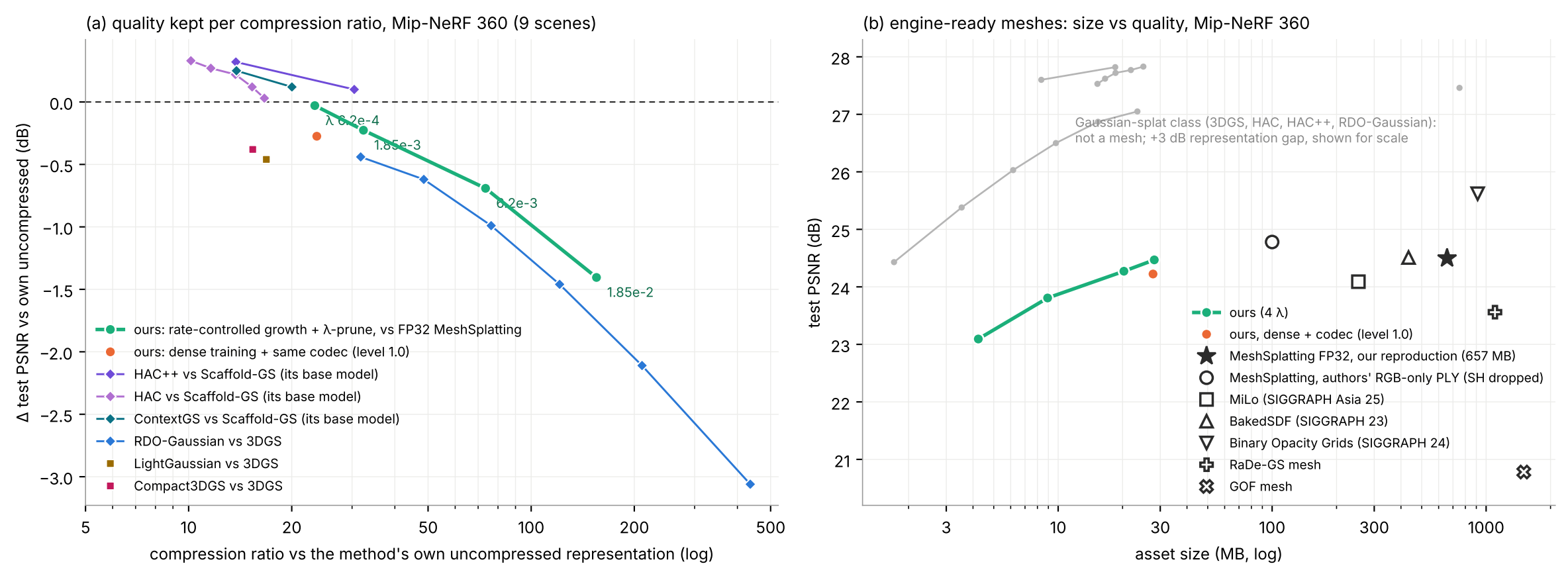
(a) quality kept per compression ratio, Mip-NeRF 360 (9 scenes)
Δ test PSNR vs own uncompressed (dB)
compression ratio vs the method's own uncompressed representation (log)
λ 6.2e-4
1.85e-3
6.2e-3
1.85e-2
ours: rate-controlled growth + λ-prune, vs FP32 MeshSplatting
ours: dense training + same codec (level 1.0)
HAC++ vs Scaffold-GS (its base model)
HAC vs Scaffold-GS (its base model)
ContextGS vs Scaffold-GS (its base model)
RDO-Gaussian vs 3DGS
LightGaussian vs 3DGS
Compact3DGS vs 3DGS
(b) engine-ready meshes: size vs quality, Mip-NeRF 360
test PSNR (dB)
asset size (MB, log)
Gaussian-splat class (3DGS, HAC, HAC++, RDO-Gaussian):
not a mesh; +3 dB representation gap, shown for scale
ours (4 λ)
ours, dense + codec (level 1.0)
MeshSplatting FP32, our reproduction (657 MB)
MeshSplatting, authors' RGB-only PLY (SH dropped)
MiLo (SIGGRAPH Asia 25)
BakedSDF (SIGGRAPH 23)
Binary Opacity Grids (SIGGRAPH 24)
RaDe-GS mesh
GOF mesh